Explainable Accounting-Fraud Detection with Machine Learning

Extending Bao, Ke, Li, Yu & Zhang (Journal of Accounting Research, 2020)

Minor project · scale-free features, firm history, tuned ensembles, decision metrics, calibrated scores and SHAP explanations

What this talk covers

1 Paper highlights

What Bao et al. (2020) did and found

2 Literature review

Data, protocol, metrics, benchmarks, limitations

3 Research gaps

Five concrete things the paper leaves open

4 Our implementation

Features, models, evaluation rigour, deployment

5 Results

All numbers, with confidence intervals, and what failed

6 Future work

Realistic major-project extensions

1 · The research paper

Bao, Ke, Li, Yu & Zhang (2020), "Detecting Accounting Fraud in Publicly Traded U.S. Firms Using a Machine Learning Approach", Journal of Accounting Research 58(1). About 330 citations.

Paper highlights and key summary

28

raw accounting items beat 14 expert ratios

0.72

AUC of RUSBoost on 2003-2008 test years (as reported, approx.)

2 yr

gap between training and test years (fraud is found late)

NDCG@k

new metric: quality of the TOP of the ranking

- Problem: predict which U.S. public firms will be charged with accounting fraud (SEC AAERs) one year ahead.
- Method: RUSBoost, boosting plus random under-sampling, built for rare events (0.66% of firm-years are fraud).
- Inputs: 28 raw Compustat items chosen from accounting theory, instead of hand-made ratios.
- Benchmarks: Dechow et al. (2011) F-score logistic model and Cecchini et al. (2010) SVM with a financial kernel.
- Finding: the ensemble on raw items outperforms both benchmarks on AUC and especially at the top of the list (NDCG@1%).
- Why it matters: a regulator or auditor can only investigate a few firms; ranking quality at the top is what counts.

Literature review: data and labels

Fraud labels (AAERs)

SEC Accounting and Auditing Enforcement Releases compiled by UC Berkeley CFRM (May 1982 to Sep 2016), extended by the authors by hand to Dec 2018 (AAER #4012). A firm-year is "fraud" if it is covered by an AAER. Serial frauds share one case id (p_aaer).

Financial data (Compustat)

Annual fundamentals, fiscal years 1991-2014. 146,045 firm-years from 18,444 firms; 964 fraud firm-years in 412 AAER cases, a base rate of 0.66%.

28 raw items

act, ap, at, ceq, che, cogs, csho, dlc, dltis, dltt, dp, ib, invt, ivao, ivst, lct, lt, ni, ppegt, pstk, re, rect, sale, sstk, txp, txt, xint, prcc_f

14 financial ratios (Dechow et al.)

dch_wc, ch_rsst, dch_rec, dch_inv, soft_assets, ch_cs, ch_cm, ch_roa, issue, bm, dpi, reoa, EBIT, ch_fcf. The paper shows raw items beat these ratios, so the hand-made features of prior work were not helping the model.

Literature review: protocol and model

1	**Expanding window**	Train on 1991 to T-2, test on year T, for T = 2003 to 2014. Mimics how a regulator would use it.
2	**2-year gap**	Fraud in year T-1 is usually not public yet, so T-1 is dropped from training. Prevents leakage of undiscovered fraud.
3	**Serial-fraud masking**	A fraud case that continues into the test year is removed from training so the model cannot "remember" the firm.
4	**Tuning**	Train 1991-1999, validate on 2001 (gap kept). Grid over number of learners 100 to 3000, chosen by AUC.
5	**RUSBoost**	300 boosted trees, MinLeafSize 5, learning rate 0.1, under-sample non-fraud to 1:1 each round (MATLAB fitensemble).

Literature review: metrics and findings

Metric	What it measures	Why the paper uses it
AUC	Probability a random fraud ranks above a random non-fraud	Standard, threshold-free
NDCG@k	Discounted hits in the top k% of the ranking	Investigators only look at the top of the list
Sensitivity@k	Share of all frauds caught in the top k%	Recall at a budget
Precision@k	Share of the top k% that are fraud	Hit rate per investigation

Limitations the authors acknowledge: only SEC-caught fraud is labelled (undetected fraud = 0); labels keep arriving for years; the model ranks, it does not decide.

Reported findings (approx., Table 3)

RUSBoost on 28 raw items: AUC about 0.72 over 2003-2008, ahead of the logistic F-score (about 0.69) and SVM-FK (about 0.63).

The raw-item model also wins on NDCG@1%, i.e. at the very top of the list.

Results are robust to a 1-year gap and to alternative test windows.

Our reproduction of the baseline: AUC 0.705 (paper tree settings) to 0.724 (deeper trees) over 2003-2014, consistent with the paper.

2 · Research gaps

What the paper leaves open, found by re-implementing it line by line

Five gaps we set out to close

1. Scale bias

Raw dollar items (assets, shares, share price) make the model partly learn "big firms that issue equity". Risk is not scale-free.

2. No explanation

RUSBoost is a black box of 300 trees. An auditor cannot see WHY a firm is flagged.

3. No decision step

The paper ranks firms. There is no cutoff, no confusion matrix, no calibrated probability an investigator can act on.

4. Thin significance

Model differences are reported without confidence intervals; fraud cases span years, so observations are not independent.

5. Not deployable

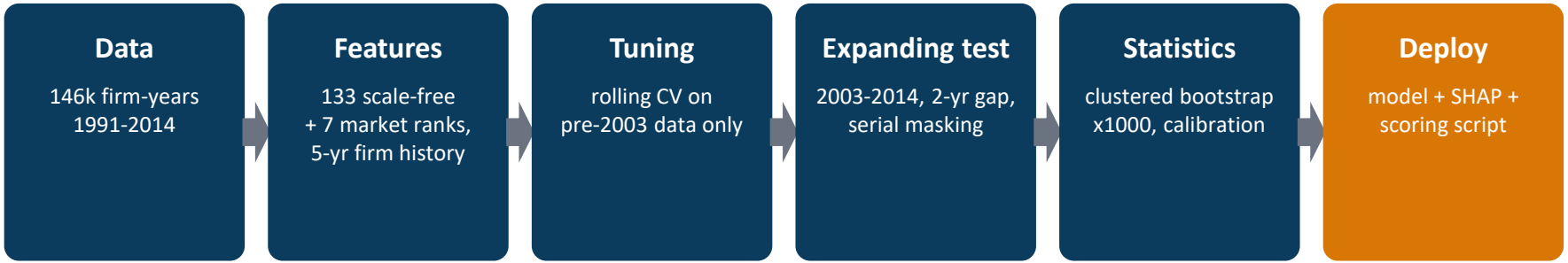
No scoring path for new firms; later training years have incomplete labels (fraud is found years later).

Each gap maps to one implemented improvement on the next slides.

3 · What we implemented

Same protocol as the paper, so every number is comparable

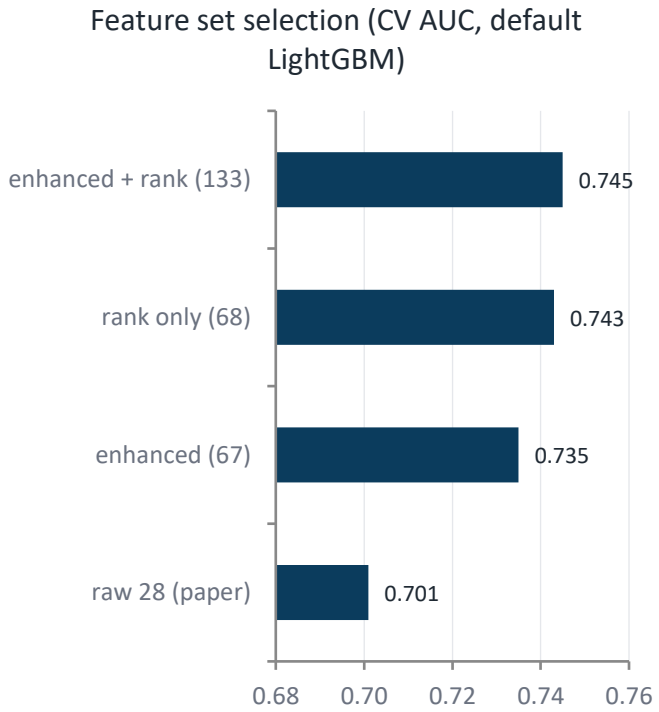
Pipeline overview



- Everything the paper did is kept: expanding window, 2-year gap, serial-fraud masking, tuning only on data before 2003.
- Added: winsorising refit on every training window (no fixed 1991-2001 caps leaking into 2014); test years with under 15 frauds excluded from averages (2013, 2014).
- Baseline re-implemented in Python from the authors' MATLAB settings (300 trees, leaf 5, LR 0.1, 1:1, 10 splits per tree) and reproduces their AUC range.
- Code: run_pipeline.py (about 2 hours on a laptop CPU), src/{features, models, deep_models, evaluate, explain}.py, score_new_data.py, smoke test.

Improvement 1: scale-free feature engineering

- Dropped all 28 raw dollar items from the proposed model (kept for the baseline only).
- Added Beneish M-score components (DSRI, GMI, AQI, SGI, DEPI, TATA, LVGI), Altman Z, common-size balance-sheet ratios, growth gaps, external-financing need.
- Added within-year percentile ranks of every ratio: a firm is compared with its own year's market, which removes era drift (1991 vs 2014).
- Feature set chosen by validation AUC on pre-2003 folds, never on test years.
- Later added 7 within-year ranks of market-value and share-count movement, tested under separate rules written in advance.



Improvement 2: tuned, balanced-bagged ensemble

- Members: LightGBM, XGBoost and logistic regression, plus a Temporal Transformer over each firm's last 5 years. Trees are balanced bags: all frauds plus a 20x sample of non-frauds, 10 bags.
- Hyper-parameters by random search on rolling-origin CV using only pre-2003 data: 20 trials, then 40 more per tree model, chosen by the full ensemble's AUC.
- Members are rank-averaged within each test year, exactly as the paper scores within a year.
- Ensemble chosen on validation AUC with a simplicity tie-break; the baseline model is never a member.

Candidate	CV AUC
RUSBoost raw 28 (baseline)	0.690
RUSBoost enhanced	0.736
LightGBM	0.746
XGBoost	0.749
Logistic regression	0.743
LGBM + XGB	0.751
LGBM + XGB + LR	0.760
+ Temporal Transformer	0.772
+ market, recency, retuned (final)	0.787

Improvements 3 to 5: decisions, statistics, deployment

3. Decision step

Precision, recall and lift at a 1%, 2% and 5% investigation budget, per year, plus a confusion matrix at 1%.

Isotonic regression turns the rank score into a calibrated fraud probability (fit on out-of-sample 2003-2010 scores).

Plain-language output: "frauds found per 100 investigations".

4. Honest statistics

Paired bootstrap, 1,000 resamples, clustered by AAER case for frauds and by firm for non-frauds (observations are not independent).

CIs for AUC, NDCG@5% and Sens@5%, not just AUC.

Years with fewer than 15 frauds (2013: 11, 2014: 4) are listed but excluded from averages.

5. Deployment and SHAP

Deployable model trained on 1991-2010 only, because 2011+ labels are still arriving (69 frauds in 2003 vs 4 in 2014).

SHAP computed on all four deployed members, on a real-base-rate sample of 2012-2014. The Temporal Transformer uses expected gradients.

score_new_data.py ranks new firms against the full market of the same year and outputs probability and percentile.

Beyond the gaps: firm history and market signals

0.735

test AUC, LGBM + XGB + LR (3 seeds)

0.743

+ Temporal Transformer

0.748

+ market features, recency weights, retuned trees

Firm history

A Temporal Transformer reads each firm's last 5 fiscal years (68 rank features, 3 networks).

Gain +0.0084, 95% CI [+0.0036, +0.0135], p = 0.003, Holm 0.11.

Market signals

7 ranks of market-value and share movement; tree rows lose half their weight every 3 years; trees retuned for the ensemble.

Gain +0.0051, 95% CI [+0.0003, +0.0103], p = 0.019, Holm 0.17.

How to read them

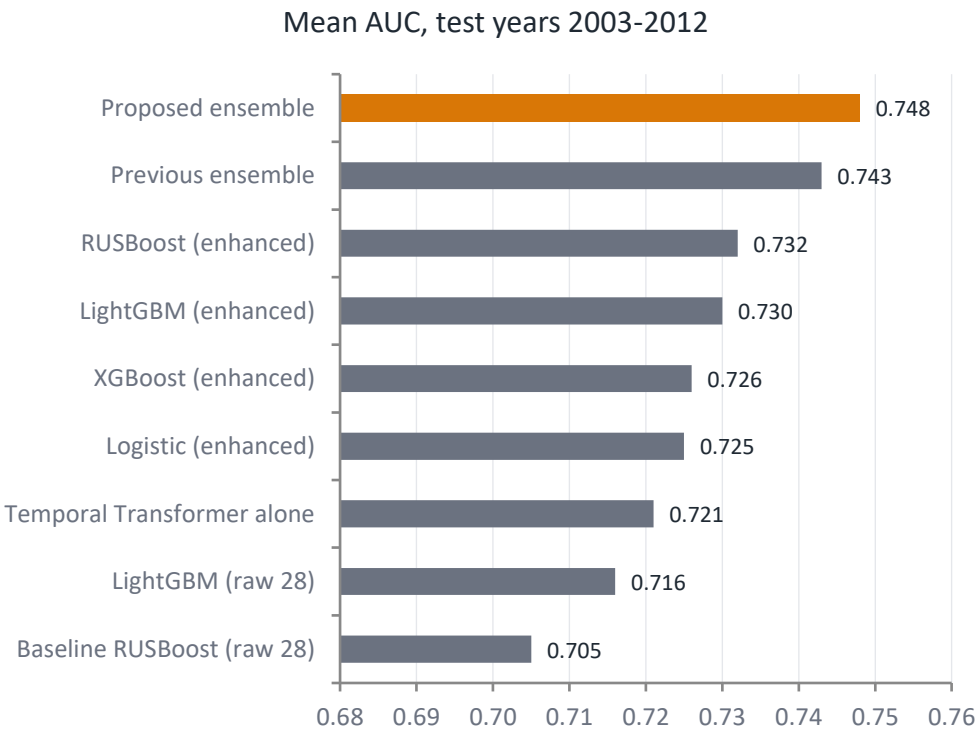
Rules written first; candidates frozen on pre-2003 data. Neither survives Holm at 0.05, so both are "promising, not confirmed".

The second is exploratory: the test years had been used before.

4 · Results

Test years 2003-2012 (2013-2014 excluded for too few frauds); mean of yearly metrics; same protocol as the paper; one pipeline run (seed 42) unless stated

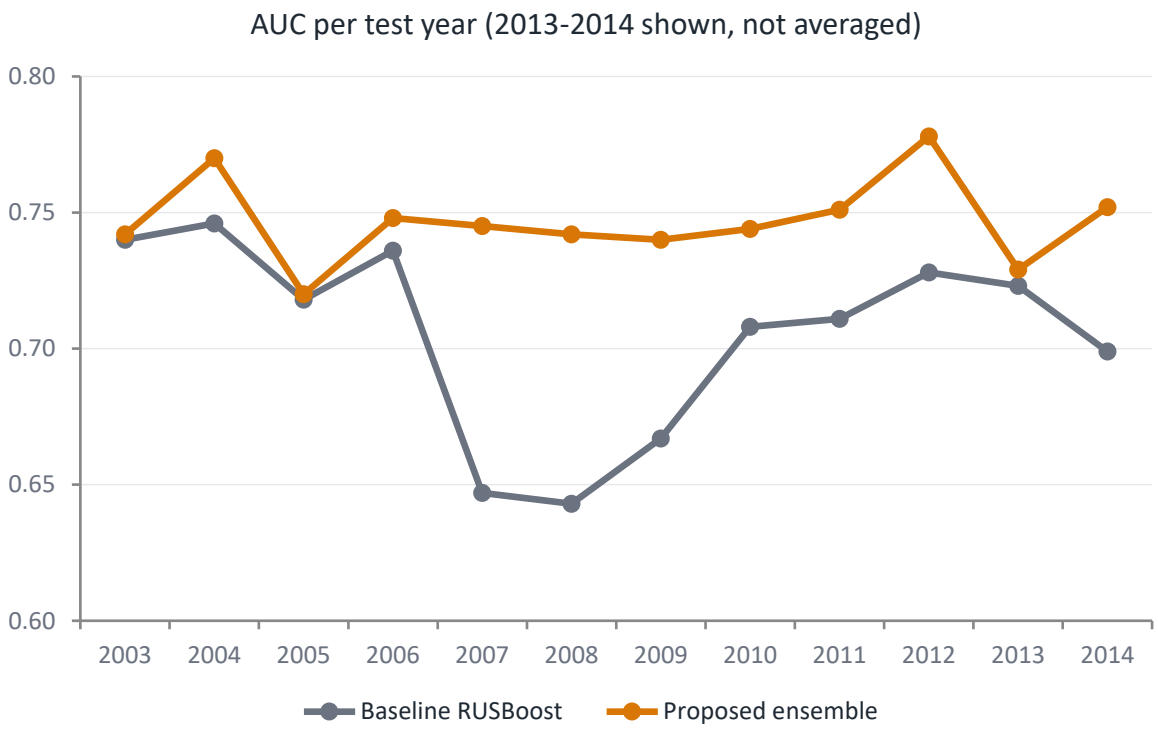
Headline: AUC 0.705 to 0.748, a significant gain



Metric	Gain	95% CI	P(gain<= 0)
AUC	+0.044	[+0.025, +0.062]	0.000
Sens@5%	+0.053	[+0.013, +0.093]	0.003
NDCG@5%	+0.015	[-0.011, +0.042]	0.118

- AUC and Sens@5% gains are statistically significant under a case-clustered bootstrap; NDCG@5% improves, not significantly.
- RUSBoost on the new features alone reaches 0.732: the features do most of the work. Ensemble, firm history and market signals add 0.016.
- Against the previous ensemble, 3 seeds: AUC 0.7433 to 0.7484, p = 0.019, Holm 0.17. Promising, not confirmed.

Year by year: ahead in all 10 scored years



- Baseline collapses in 2007-2009 (AUC 0.64-0.67, the financial-crisis years); the proposed model stays near 0.74.
- Proposed wins all 10 scored years. The smallest margins are 2003 and 2005, about 0.002 each.
- Fraud counts per test year fall from 69 to 4, which is why late years are noisy and why the deployable model stops training at 2010.

Decision step: what an investigator actually gets

1.9
frauds per 100 investigations (top 1%), proposed

1.8
baseline RUSBoost

0.6
random picking (base rate)

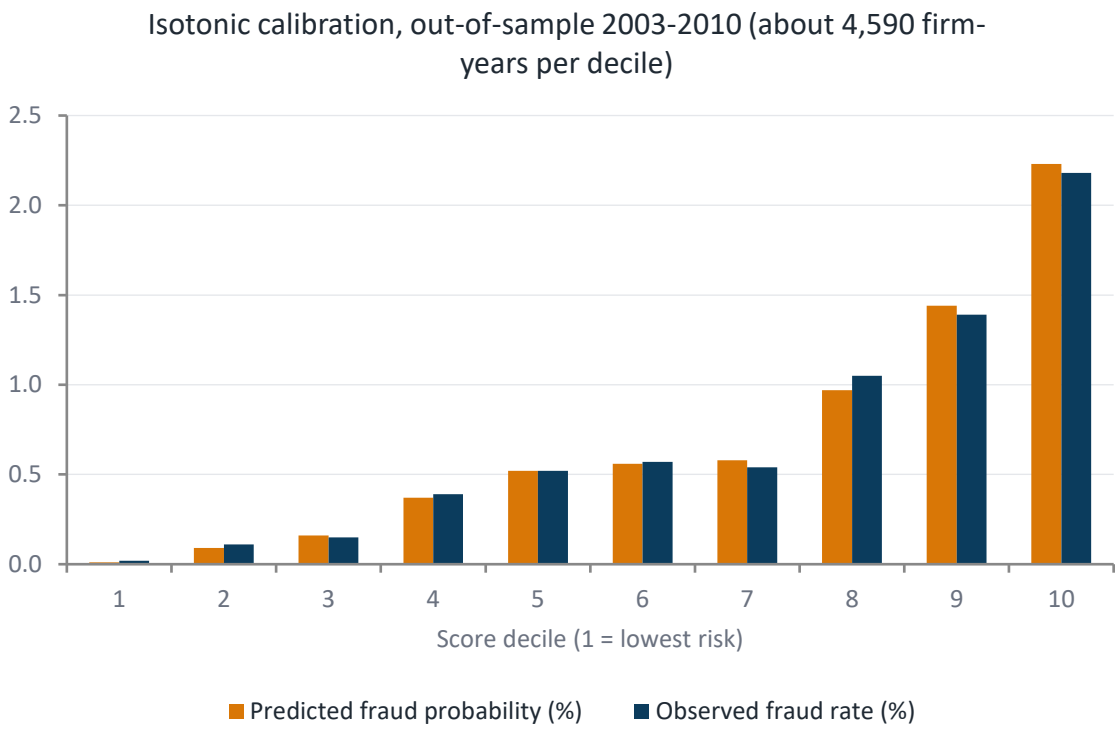
2.7x
mean yearly lift over random

Budget	Precision	Recall	Lift
Top 1%	1.9%	2.7%	2.7x
Top 2%	1.8%	5.6%	2.8x
Top 5%	2.0%	15.1%	3.0x

2004 (best year), top 1%	Fraud	Not fraud
Flagged	TP 3	FP 57
Not flagged	FN 55	TN 5,819

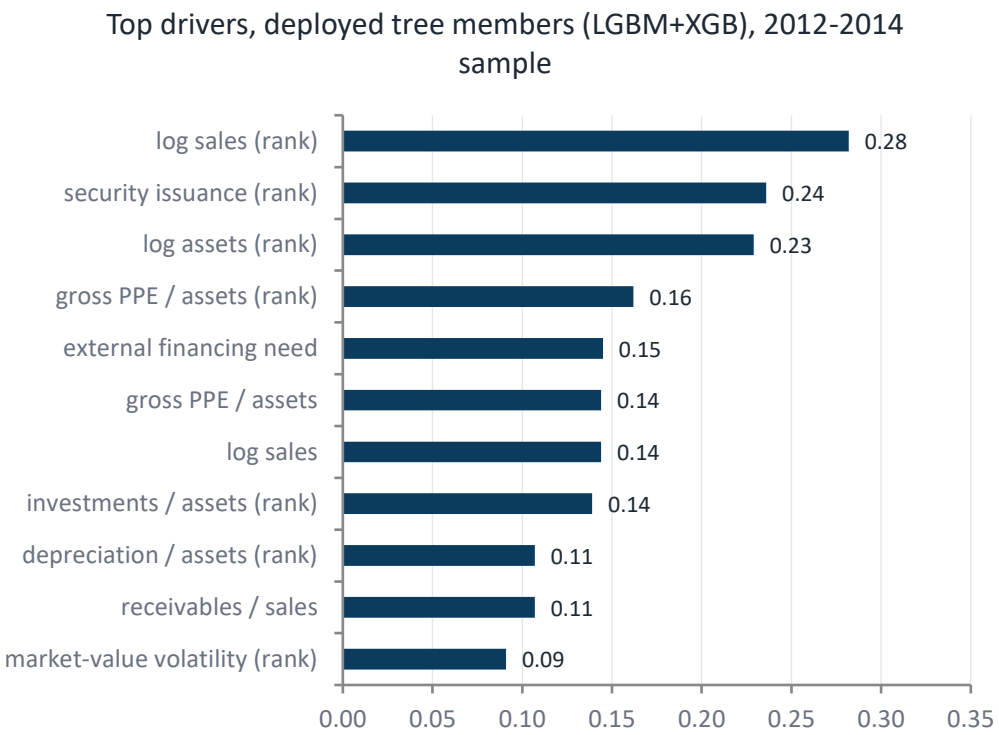
Honest framing: about 98 of every 100 flagged firms are not (labelled) fraud. This is a prioritisation tool that raises an investigator's hit rate about 2.7x, not a fraud detector. Three years (2008, 2009, 2012) have zero hits at 1%. The last upgrade helped overall AUC, not this budget: over 3 seeds, Prec@1% fell from 2.1% to 1.9%.

Calibrated probabilities: the score means what it says



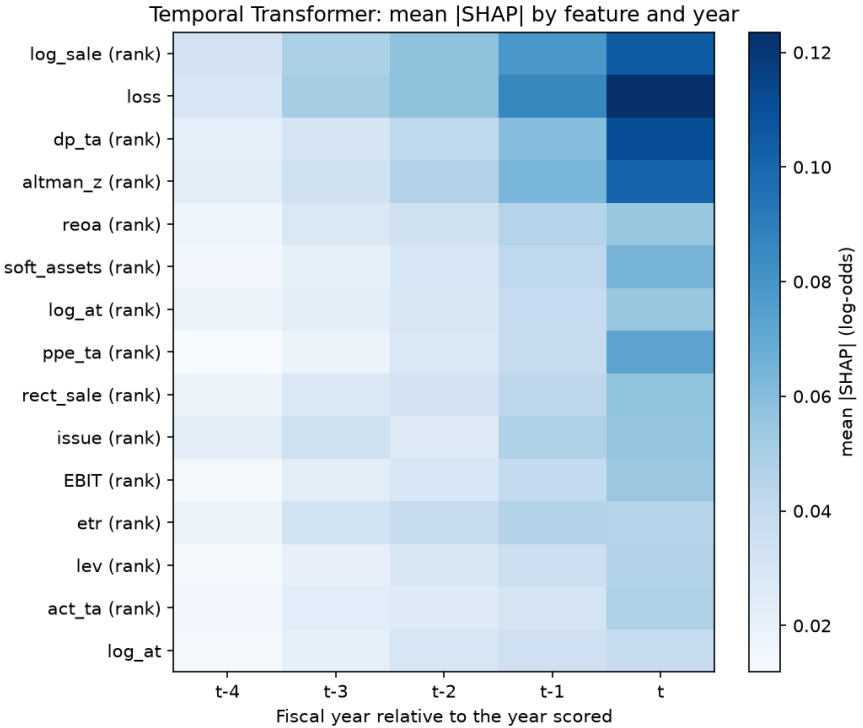
- Predicted and observed rates are close in every decile; top decile 2.2% predicted vs 2.2% observed.
- Bagging makes raw model outputs far higher than real fraud rates; calibration fixes that before anyone reads them as probabilities.
- Limits: the calibrator is fit on these same scores, so close agreement is partly by construction. It is a step function fit on 8 years and saturates near 2-3%.

Explainability: SHAP on the deployed model



- Drivers an auditor recognises: security issuance, external financing need, heavy fixed assets, investments, receivables / sales. Market-value volatility is the strongest new market feature, 11th overall.
- A flagged firm can be opened as a waterfall plot: which features pushed its score up or down.
- Honest caveat: firm size (log sales, log assets) still dominates even without raw dollar items. A size-neutral model has not been tested.
- SHAP shows what the model uses, not why a firm committed fraud; the top-flagged table shows real hits and false alarms side by side.

Explainability: the Temporal Transformer



- Explained directly with expected gradients, the method behind SHAP's GradientExplainer: the same 3,000 firm-years, in the same log-odds units as the trees.
- Its top drivers include the rank of sales, the loss flag, and the ranks of depreciation, Altman Z and retained earnings. Five of its top 10 are also in the trees' top 10. This ranking is indicative: see the next slide.
- It weights recent years most: 38% of attribution on the latest year, 9% on t-4. Older years exist for fewer firm-years (t-4 for 58%), but the slope remains among those that have them.
- Any flagged firm-year can be opened as a waterfall, with all four member ranks beside its drivers.

Can the explanations be trusted? 7 of 8 checks passed

Check, limits fixed before any result	Result	Limit	Verdict
A Known answer on a linear network	error 4.8e-07	below 1e-4	Pass
B Same output as the deployed network; same ranking	gap 9.5e-07; 0.997	below 1e-4; 0.99	Pass
C Attributions sum to the output (median; 99th pct, in SDs)	0.004; 0.020	0.01; 0.05	Pass
D Second method: global; within firm-year; year-share gap	0.986; 0.974; 0.037	0.80; 0.80; 0.05	Pass
E Stable across references; across the 3 networks	0.984; 0.806	0.95; 0.70	Pass
F Resetting top 20 cells lowers the logit, vs 20 random	0.962 vs 0.018	2 times random	Pass
G Global ranking vs re-initialised weights; shuffled labels	**0.425; 0.179**	**0.30; 0.50**	**FAIL**
H Same result on a repeat run	difference 0.0	below 1e-3	Pass

What failed

An untrained network shares part of the global ordering (0.35 on average over 10). That ranking is indicative only. The cause is not established.

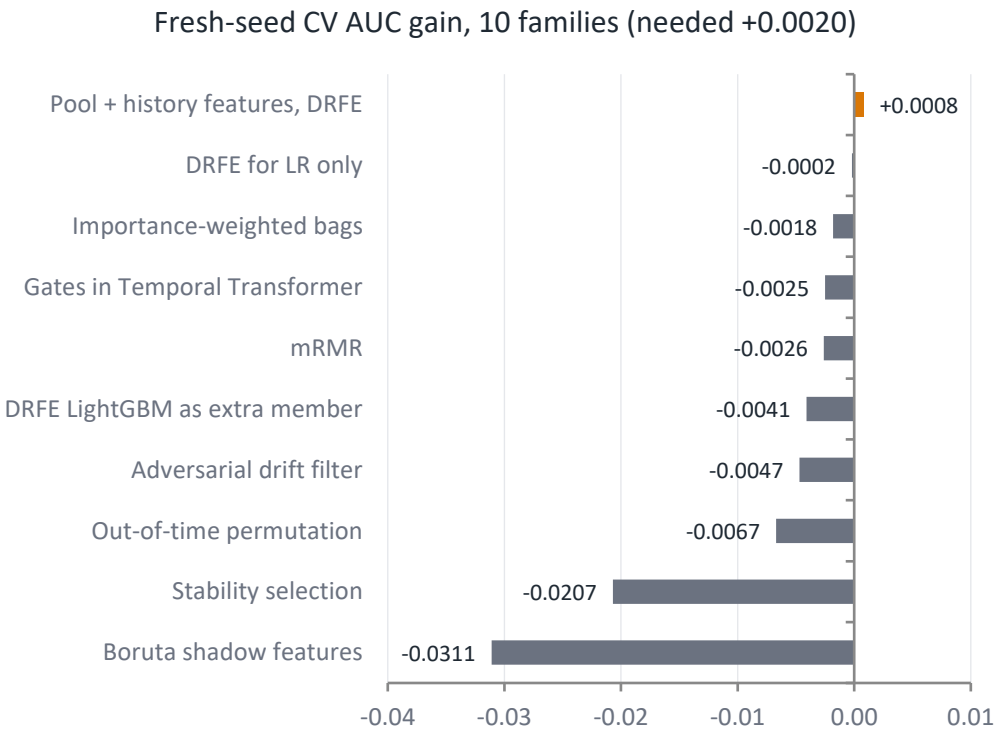
What holds

Within a firm-year, random networks are unrelated to the real one (−0.03). Resetting real top cells lowers the logit by 0.96; a random network's raise it by 0.09.

What changed

Model scores, metrics and tree SHAP are identical after the rerun. The follow-up was chosen after the failure, so it cannot overturn it.

Feature selection and generation: tested, no gain



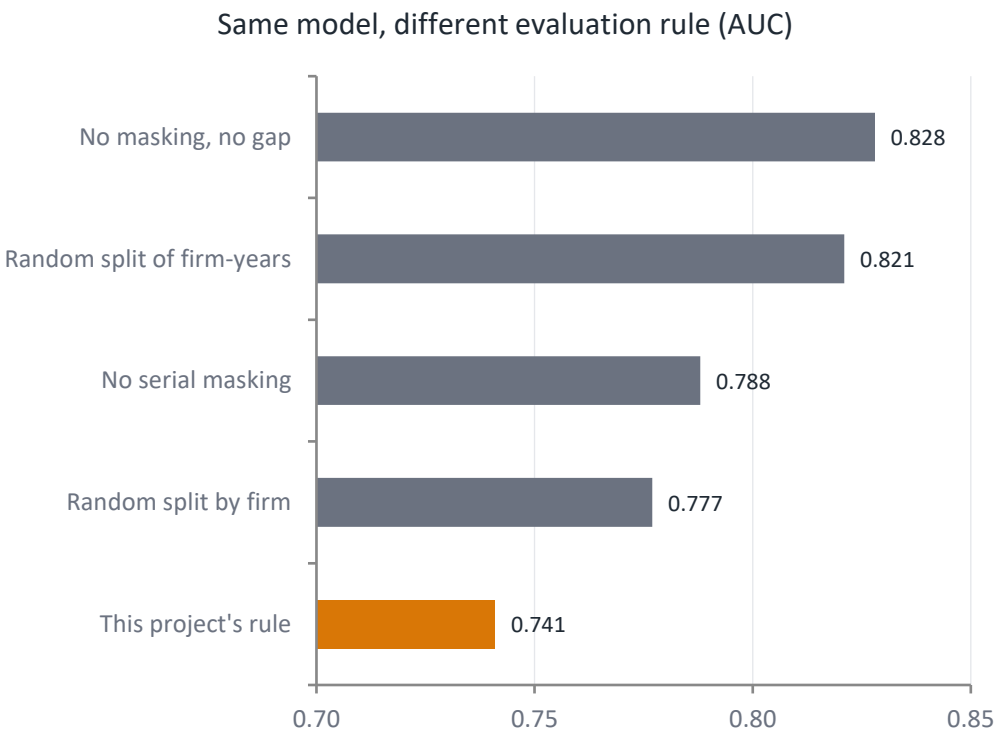
- The 133 features are a fixed pool, chosen as a set on validation AUC. Why not select inside each training window?
- 10 selection families ran inside every window, under rules written in advance. Each needed +0.002 CV AUC on fresh seeds; the best reached +0.0008.
- Is it at least no worse? On the test years, dynamic selection changed AUC by +0.0008 (95% CI −0.0043 to +0.0062). Inconclusive against a 0.003 margin, so the pool stays fixed.
- Automatic feature generation (OpenFE-style) gained −0.0009 and −0.0018 on fresh seeds, so it never saw the test years.
- These are the families tried, not every method that exists.

Other candidates: what did not make the model

Candidate	Ref. AUC	CV gain	Test AUC	Verdict
FT-Transformer added	0.735	-0.0036	0.728	Rejected
TabM added	0.735	-0.0026	0.726	Rejected
Mambular added	0.735	-0.0011	0.729	Rejected
Dynamic RFE	0.735	+0.0003	0.733	Rejected
Focal PU XGBoost replacing XGBoost	0.735	+0.0005	0.731	Rejected
Temporal Mamba added	0.735	+0.0041	0.738	Passed, weaker; not used
Adopted + peer deviations + TT x 3	0.743	+0.0149	0.7482	Rejected, p = 0.078
Adopted + TT x 3	0.743	+0.0141	0.7476	Rejected, p = 0.095
Filing text, industry, timing, amendments	0.743	best +0.0014	not run	Failed validation gate
8 other families (accruals, pairwise loss...)	0.743	best +0.0013	not run	Failed validation gate

Ref. AUC is the test AUC of the model each candidate had to beat. CV gain is on validation years. The code for what failed stays in git history.

Why some papers report above 0.8



- Same trees and logistic regression, seed 42, without the Temporal Transformer. Only the evaluation changes.
- Dropping serial-fraud masking adds 0.047; also dropping the 2-year gap adds another 0.040.
- Weaker rules keep labels a real user would not have yet, because AAERs arrive years later. A random split of firm-years also puts one fraud case on both sides.
- Our test AUC, 0.748, stays below 0.8. These rows are a diagnostic and choose nothing.
- Random splits cover 1991-2012, so compare them loosely.

Confident lines to present

"We reproduced the JAR 2020 baseline in Python with the authors' exact settings, and it lands in their reported AUC range."

"Scale-free features, firm history and market signals lift AUC from 0.705 to 0.748 and top-5% recall from 9.8% to 15.1%, both significant under a case-clustered bootstrap."

"Unlike the paper, we give an investigator a decision: at a 1% budget the model finds 1.9 frauds per 100 reviews, 2.7x random on average, with a calibrated probability behind every score."

"Flagged firms come with a SHAP explanation from each deployed member, not a proxy model. The Temporal Transformer's passed 7 of 8 audit checks; its global ranking failed one and is indicative only."

"The deployable model stops at 2010 on purpose: fraud labels for later years are still being discovered."

"We did not reach 0.8 and will not claim it: the same trees and LR score 0.82-0.83 only under weaker rules that use labels a real user would not have yet."

5 · What a major project could add

Suggestions only, none implemented. Ordered by expected value per week of work.

Realistic extensions for a major project

Audit and governance data

Audit fees, auditor changes, board data and stock returns (a WRDS subscription is needed). Bertomeu et al. (2021) find such variables matter. Filing text and tone were tested here and failed their gate.

Size-neutral model

Drop log assets / log sales, or model fraud risk within size buckets, and test whether precision at 1% holds. Decides whether "size" is signal or bias.

Cost-sensitive cutoff

Pick the investigation budget by expected value: cost of a review vs. loss avoided per catch. Turns the ranking into a defendable policy.

Label-delay modelling

Weight recent training years by the probability their frauds are already discovered (survival-style), instead of a hard 2010 cutoff.

Stronger statistics

DeLong test per year, firm-level cross-validated calibration, and a stricter nested selection so the 3 validation years cannot overfit.

Other markets

Re-run on an Indian or EU sample (different standards, different fraud patterns).

Thank you

Code, results and this deck: Final-Year-Project repository

run_pipeline.py reproduces the main results in about 2 hours on a laptop; score_new_data.py scores new firms.